

Warmup Assignment — nanoGPT on Shakespeare

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1. Environment & Node Allocation

System: NERSC Perlmutter (GPU). Do not run on login nodes.

```
srun -A bcrc-dtai-gh --qos=bcrc-dtai-gh -p ghx4 -t 4:00:00 --nodes=1 --ntasks-per-node=1 --cpus-per-task=16 --gres=gpu:h100:1 --mem=256g
--pty bash -i
module load cuda/12.6.1
module load gcc/11.4.0
```

2. Installation Commands

```
git clone https://github.com/karpathy/nanoGPT
cd nanoGPT/
conda create -n nanogpt python=3.10
conda activate nanogpt
pip install torch --index-url https://download.pytorch.org/whl/cu126
pip install numpy transformers datasets tiktoken wandb tqdm
```

3. Data Processing Commands

```
python data/shakespeare_char/prepare.py
```

4. Training Commands

```
python train.py config/train_shakespeare_char.py
```

5. Representative Logs

Overriding config with config/train_shakespeare_char.py:

```
# train a miniature character-level shakespeare model
# good for debugging and playing on macbooks and such
```

```
out_dir = 'out-shakespeare-char'
eval_interval = 250 # keep frequent because we'll overfit
eval_iters = 200
log_interval = 10 # don't print too too often
# we expect to overfit on this small dataset, so only save when val improves
```

```

always_save_checkpoint = False

wandb_log = False # override via command line if you like
wandb_project = 'shakespeare-char'
wandb_run_name = 'mini-gpt'

dataset = 'shakespeare_char'
gradient_accumulation_steps = 1
batch_size = 64
block_size = 256 # context of up to 256 previous characters

# baby GPT model :)
n_layer = 6
n_head = 6
n_embd = 384
dropout = 0.2

learning_rate = 1e-3 # with baby networks can afford to go a bit higher
max_iters = 5000
lr_decay_iters = 5000 # make equal to max_iters usually
min_lr = 1e-4 # learning_rate / 10 usually
beta2 = 0.99 # make a bit bigger because number of tokens per iter is small

warmup_iters = 100 # not super necessary potentially

# on macbook also add
# device = 'cpu' # run on cpu only
compile = False # do not torch compile the model

tokens per iteration will be: 16,384
found vocab_size = 65 (inside data/shakespeare_char/meta.pkl)
Initializing a new model from scratch
number of parameters: 10.65M
/u/ddong/workspace/nanoGPT/train.py:196: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use
`torch.amp.GradScaler('cuda', args...)` instead.
  scaler = torch.cuda.amp.GradScaler(enabled=(dtype == 'float16'))
num decayed parameter tensors: 26, with 10,740,096 parameters
num non-decayed parameter tensors: 13, with 4,992 parameters
using fused AdamW: True
step 0: train loss 4.2874, val loss 4.2823
iter 0: loss 4.2714, time 3520.31ms, mfu -100.00%
iter 10: loss 3.1476, time 12.42ms, mfu 30.00%
iter 20: loss 2.7333, time 12.17ms, mfu 30.06%
iter 30: loss 2.6157, time 12.02ms, mfu 30.15%

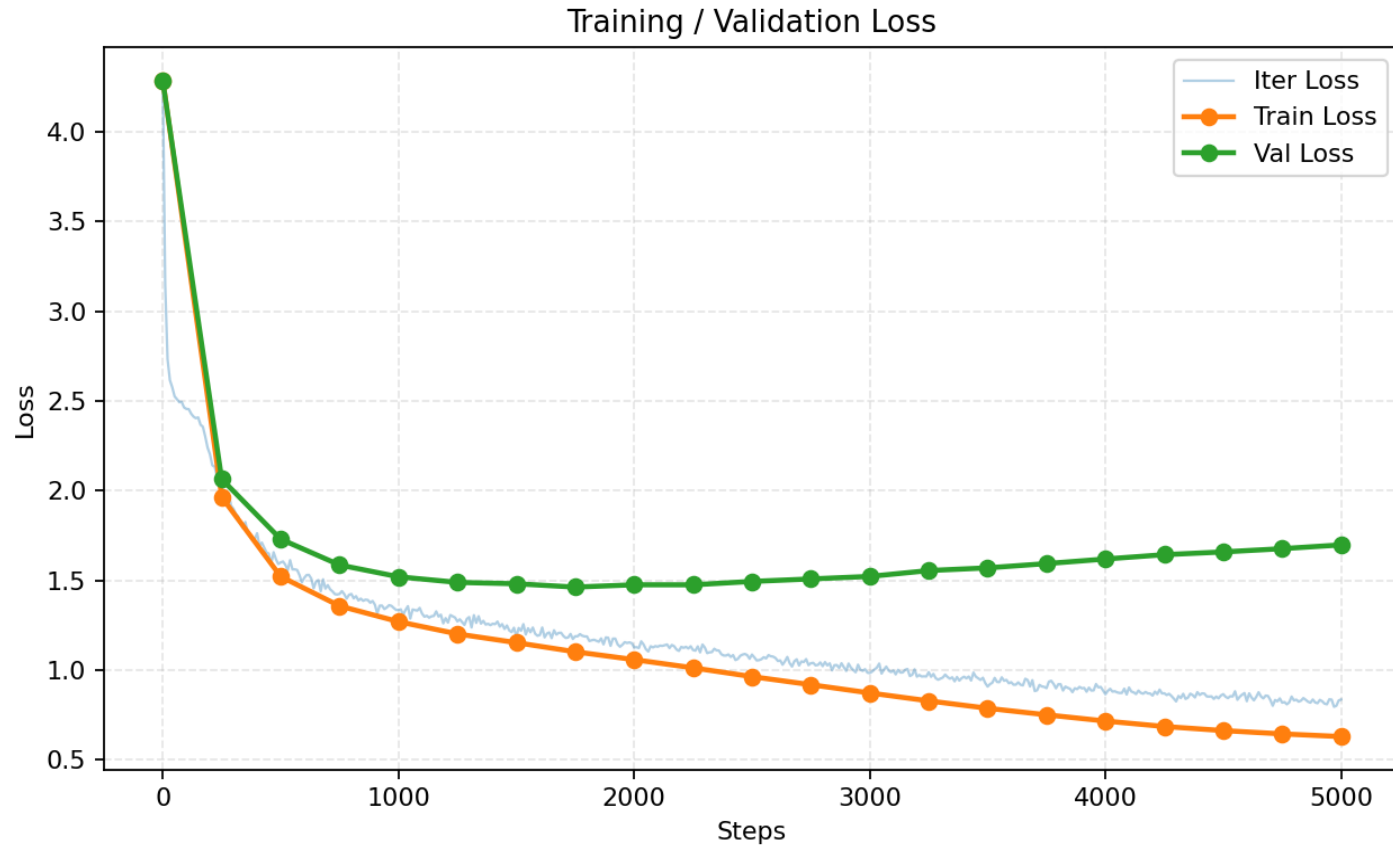
```

```
iter 40: loss 2.5761, time 12.08ms, mfu 30.22%
iter 50: loss 2.5257, time 12.19ms, mfu 30.26%
iter 60: loss 2.5101, time 12.18ms, mfu 30.29%
iter 70: loss 2.4937, time 12.17ms, mfu 30.33%
iter 80: loss 2.4949, time 12.14ms, mfu 30.36%
iter 90: loss 2.4642, time 12.29ms, mfu 30.36%
iter 100: loss 2.4559, time 12.06ms, mfu 30.41%
iter 110: loss 2.4545, time 12.18ms, mfu 30.43%
iter 120: loss 2.4294, time 12.05ms, mfu 30.48%
iter 130: loss 2.4133, time 12.24ms, mfu 30.48%
iter 140: loss 2.4042, time 12.19ms, mfu 30.49%
iter 150: loss 2.4085, time 12.06ms, mfu 30.53%
iter 160: loss 2.3676, time 12.12ms, mfu 30.55%
iter 170: loss 2.3564, time 12.33ms, mfu 30.52%
iter 180: loss 2.3013, time 12.19ms, mfu 30.52%
iter 190: loss 2.2393, time 12.17ms, mfu 30.53%
iter 200: loss 2.2029, time 11.99ms, mfu 30.59%
iter 210: loss 2.1412, time 12.04ms, mfu 30.62%
iter 220: loss 2.1340, time 12.06ms, mfu 30.65%
iter 230: loss 2.0779, time 12.15ms, mfu 30.65%
iter 240: loss 2.0766, time 12.06ms, mfu 30.68%
step 250: train loss 1.9638, val loss 2.0642
saving checkpoint to out-shakespeare-char
iter 250: loss 2.0311, time 2550.40ms, mfu 27.62%
iter 260: loss 1.9738, time 12.18ms, mfu 27.92%
...
iter 4830: loss 0.8300, time 12.18ms, mfu 28.88%
iter 4840: loss 0.8442, time 12.40ms, mfu 29.00%
iter 4850: loss 0.8345, time 12.39ms, mfu 29.11%
iter 4860: loss 0.8237, time 12.03ms, mfu 29.29%
iter 4870: loss 0.8187, time 12.16ms, mfu 29.43%
iter 4880: loss 0.8363, time 12.19ms, mfu 29.54%
iter 4890: loss 0.8198, time 12.29ms, mfu 29.62%
iter 4900: loss 0.8156, time 12.33ms, mfu 29.68%
iter 4910: loss 0.8341, time 12.32ms, mfu 29.74%
iter 4920: loss 0.8262, time 12.44ms, mfu 29.76%
iter 4930: loss 0.8130, time 12.22ms, mfu 29.83%
iter 4940: loss 0.8096, time 12.42ms, mfu 29.85%
iter 4950: loss 0.8373, time 12.45ms, mfu 29.86%
iter 4960: loss 0.8353, time 12.52ms, mfu 29.85%
iter 4970: loss 0.7957, time 12.34ms, mfu 29.88%
iter 4980: loss 0.8014, time 12.24ms, mfu 29.94%
iter 4990: loss 0.8311, time 12.33ms, mfu 29.97%
step 5000: train loss 0.6296, val loss 1.6980
```

iter 5000: loss 0.8360, time 2260.00ms, mfu 26.99%

6. Loss Curves (Train & Val)

Figure 1: miniGPT-shakespeare.png (Axes: Steps vs Loss)



7. Full Command History

```
(base) ddong@gh-login03:~/workspace> git clone https://github.com/karpathy/nanoGPT
Cloning into 'nanoGPT'...
remote: Enumerating objects: 686, done.
```

```
remote: Total 686 (delta 0), reused 0 (delta 0), pack-reused 686 (from 1)
Receiving objects: 100% (686/686), 974.05 KiB | 9.55 MiB/s, done.
Resolving deltas: 100% (380/380), done.
(base) ddong@gh-login03:~/workspace> cd nanoGPT/
(base) ddong@gh-login03:~/workspace/nanoGPT> ls
█[1m█[34massets█[0m█[39m█[49m bench.py █[1m█[34mconfig█[0m█[39m█[49m configurator.py █[1m█[34mdata█[0m█[39m█[49m LICENSE model.py
  README.md sample.py scaling_laws.ipynb train.py transformer_sizing.ipynb
(base) ddong@gh-login03:~/workspace/nanoGPT> conda create -n nanogpt python=3.10
█[1m█[32m2 channel Terms of Service accepted
█[0m█[39m█[49mRetrieving notices: done
Channels:
- defaults
Platform: linux-aarch64
Collecting package metadata (repodata.json): done
Solving environment: done
```

Package Plan

```
environment location: /u/ddong/miniconda3/envs/nanogpt

added / updated specs:
- python=3.10
```

The following packages will be downloaded:

package	build	
ca-certificates-2025.9.9	hd43f75c_0	127 KB
libzlib-1.3.1	h998d150_0	65 KB
python-3.10.18	hbb0f47a_0	13.1 MB
setuptools-78.1.1	py310hd43f75c_0	1.6 MB
wheel-0.45.1	py310hd43f75c_0	115 KB
zlib-1.3.1	h998d150_0	103 KB
Total:		15.2 MB

The following NEW packages will be INSTALLED:

```
_libgcc_mutex      pkgs/main/linux-aarch64::_libgcc_mutex-0.1-main
_openmp_mutex      pkgs/main/linux-aarch64::_openmp_mutex-5.1-51_gnu
bzip2              pkgs/main/linux-aarch64::bzip2-1.0.8-h998d150_6
ca-certificates    pkgs/main/linux-aarch64::ca-certificates-2025.9.9-hd43f75c_0
expat              pkgs/main/linux-aarch64::expat-2.7.1-h419075a_0
```

ld_impl_linux-aar~	pkgs/main/linux-aarch64::ld_impl_linux-aarch64-2.40-h48e3ba3_0
libffi	pkgs/main/linux-aarch64::libffi-3.4.4-h419075a_1
libgcc-ng	pkgs/main/linux-aarch64::libgcc-ng-11.2.0-h1234567_1
libgomp	pkgs/main/linux-aarch64::libgomp-11.2.0-h1234567_1
libstdcxx-ng	pkgs/main/linux-aarch64::libstdcxx-ng-11.2.0-h1234567_1
libuuid	pkgs/main/linux-aarch64::libuuid-1.41.5-h998d150_0
libxcb	pkgs/main/linux-aarch64::libxcb-1.17.0-hf66535e_0
libzlib	pkgs/main/linux-aarch64::libzlib-1.3.1-h998d150_0
ncurses	pkgs/main/linux-aarch64::ncurses-6.5-h419075a_0
openssl	pkgs/main/linux-aarch64::openssl-3.0.17-h998d150_0
pip	pkgs/main/noarch::pip-25.2-pyhc872135_0
pthread-stubs	pkgs/main/linux-aarch64::pthread-stubs-0.3-hfd63f10_1
python	pkgs/main/linux-aarch64::python-3.10.18-hbb0f47a_0
readline	pkgs/main/linux-aarch64::readline-8.3-h886d1d0_0
setuptools	pkgs/main/linux-aarch64::setuptools-78.1.1-py310hd43f75c_0
sqlite	pkgs/main/linux-aarch64::sqlite-3.50.2-h998d150_1
tk	pkgs/main/linux-aarch64::tk-8.6.15-h987d8db_0
tzdata	pkgs/main/noarch::tzdata-2025b-h04d1e81_0
wheel	pkgs/main/linux-aarch64::wheel-0.45.1-py310hd43f75c_0
xorg-libx11	pkgs/main/linux-aarch64::xorg-libx11-1.8.12-hf66535e_1
xorg-libxau	pkgs/main/linux-aarch64::xorg-libxau-1.0.12-hf66535e_0
xorg-libxdmcp	pkgs/main/linux-aarch64::xorg-libxdmcp-1.1.5-hf66535e_0
xorg-xorgproto	pkgs/main/linux-aarch64::xorg-xorgproto-2024.1-h998d150_1
xz	pkgs/main/linux-aarch64::xz-5.6.4-h998d150_1
zlib	pkgs/main/linux-aarch64::zlib-1.3.1-h998d150_0

Proceed ([y]/n)? y

Downloading and Extracting Packages:

Preparing transaction: done

Verifying transaction: done

Executing transaction: done

#

To activate this environment, use

#

\$ conda activate nanogpt

#

To deactivate an active environment, use

#

\$ conda deactivate

(base) ddong@gh-login03:~/workspace/nanoGPT> conda activate nanogpt

The following have been reloaded with a version change:

- 1) gcc-native/13.2 => gcc-native/12.3

Lmod is automatically replacing "gcc-native/12.3" with "gcc/11.4.0".

1) cray-libsci 2) cray-mpich

```
Looking in indexes: https://download.pytorch.org/whl/cu126
```

Downloading https://download.pytorch.org/whl/cu126/torch-2.6.0%2Bcu126-cp310-cp310-linux_aarch64.whl.metadata (26 kB)

Using cached <https://download.pytorch.org/whl/filelock-3.13.1-py3-none-any.whl> (2.8 kB)

Using cached https://download.pytorch.org/whl/typing_extensions-4.12.2-py3-none-any.whl.metadata (3.0 kB)

```
Using cached https://download.pytorch.org/whl/sympy-1.13.1-py3-none-any.whl (6.2 MB)
```

Using cached <https://download.pytorch.org/whl/networkx-3.3-py3-none-any.whl>.metadata (5.1 kB)

Using cached <https://download.pytorch.org/whl/Jinja2-3.1.4-py3-none-any.whl> (2.6 kB)

Using cached <https://download.pytorch.org/whl/fsspec-2024.6.1-py3-none-any.whl>.metadata (11 kB)

```
Using cached https://download.pytorch.org/whl/mpmath-1.3.0-py3-none-any.whl (536 kB)
```

Downloading https://download.pytorch.org/whl/MarkupSafe-2.1.5-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (26 kB)

[illegible]

```
Using cached https://download.pytorch.org/whl/filelock-3.13.1-py3-none-any.whl (11 kB)
```

```
Using cached https://download.pytorch.org/whl/fsspec-2024.6.1-py3-none-any.whl (177 kB)
```

```
Using cached https://download.pytorch.org/whl/Jinja2-3.1.4-py3-none-any.whl (133 kB)
Using cached https://download.pytorch.org/whl/networkx-3.3-py3-none-any.whl (1.7 MB)
Installing collected packages: mpmath, typing-extensions, sympy, networkx, MarkupSafe, fsspec, filelock, jinja2, torch
Successfully installed MarkupSafe-2.1.5 filelock-3.13.1 fsspec-2024.6.1 jinja2-3.1.4 mpmath-1.3.0 networkx-3.3 sympy-1.13.1
torch-2.6.0+cu126 typing-extensions-4.12.2
(nanogpt) ddong@gh-login03:~/workspace/nanoGPT> pip install numpy transformers datasets tiktoken wandb tqdm
Collecting numpy
  Downloading numpy-2.2.6-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (63 kB)
Collecting transformers
  Downloading transformers-4.56.2-py3-none-any.whl.metadata (40 kB)
Collecting datasets
  Downloading datasets-4.1.1-py3-none-any.whl.metadata (18 kB)
Collecting tiktoken
  Downloading tiktoken-0.11.0-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (6.7 kB)
Collecting wandb
  Downloading wandb-0.22.0-py3-none-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (10 kB)
Collecting tqdm
  Using cached tqdm-4.67.1-py3-none-any.whl.metadata (57 kB)
Requirement already satisfied: filelock in /u/ddong/miniconda3/envs/nanogpt/lib/python3.10/site-packages (from transformers) (3.13.1)
Collecting huggingface-hub<1.0,>=0.34.0 (from transformers)
  Downloading huggingface_hub-0.35.0-py3-none-any.whl.metadata (14 kB)
Collecting packaging>=20.0 (from transformers)
  Using cached packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pyyaml>=5.1 (from transformers)
  Downloading PyYAML-6.0.2-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (2.1 kB)
Collecting regex!=2019.12.17 (from transformers)
  Downloading regex-2025.9.18-cp310-cp310-manylinux2014_aarch64.manylinux_2_17_aarch64.manylinux_2_28_aarch64.whl.metadata (40 kB)
Collecting requests (from transformers)
  Using cached requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting tokenizers<=0.23.0,>=0.22.0 (from transformers)
  Downloading tokenizers-0.22.1-cp39-abi3-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (6.8 kB)
Collecting safetensors>=0.4.3 (from transformers)
  Downloading safetensors-0.6.2-cp38-abi3-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (4.1 kB)
Requirement already satisfied: fsspec>=2023.5.0 in /u/ddong/miniconda3/envs/nanogpt/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.34.0->transformers) (2024.6.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /u/ddong/miniconda3/envs/nanogpt/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.34.0->transformers) (4.12.2)
Collecting hf-xet<2.0.0,>=1.1.3 (from huggingface-hub<1.0,>=0.34.0->transformers)
  Downloading hf_xet-1.1.10-cp37-abi3-manylinux_2_28_aarch64.whl.metadata (4.7 kB)
Collecting pyarrow>=21.0.0 (from datasets)
  Downloading pyarrow-21.0.0-cp310-cp310-manylinux_2_28_aarch64.whl.metadata (3.3 kB)
Collecting dill<0.4.1,>=0.3.0 (from datasets)
  Downloading dill-0.4.0-py3-none-any.whl.metadata (10 kB)
Collecting pandas (from datasets)
```


Downloading pandas-2.3.2-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (91 kB)
Collecting xxhash (from datasets)
Downloading xxhash-3.5.0-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (12 kB)
Collecting multiprocessing<0.70.17 (from datasets)
Downloading multiprocessing-0.70.16-py310-none-any.whl.metadata (7.2 kB)
Collecting aiohttp!=4.0.0a0,!4.0.0a1 (from fsspec[http]<=2025.9.0,>=2023.1.0->datasets)
Downloading aiohttp-3.12.15-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (7.7 kB)
Collecting click>=8.0.1 (from wandb)
Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting gitpython!=3.1.29,>=1.0.0 (from wandb)
Using cached gitpython-3.1.45-py3-none-any.whl.metadata (13 kB)
Collecting platformdirs (from wandb)
Using cached platformdirs-4.4.0-py3-none-any.whl.metadata (12 kB)
Collecting protobuf!=4.21.0,!5.28.0,<7,>=3.19.0 (from wandb)
Downloading protobuf-6.32.1-cp39-abi3-manylinux2014_aarch64.whl.metadata (593 bytes)
Collecting pydantic<3 (from wandb)
Downloading pydantic-2.11.9-py3-none-any.whl.metadata (68 kB)
Collecting sentry-sdk>=2.0.0 (from wandb)
Downloading sentry_sdk-2.38.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting annotated-types>=0.6.0 (from pydantic<3->wandb)
Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.33.2 (from pydantic<3->wandb)
Downloading pydantic_core-2.33.2-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (6.8 kB)
Collecting typing-inspection>=0.4.0 (from pydantic<3->wandb)
Using cached typing_inspection-0.4.1-py3-none-any.whl.metadata (2.6 kB)
Collecting charset_normalizer<4,>=2 (from requests->transformers)
Downloading charset_normalizer-3.4.3-cp310-cp310-manylinux2014_aarch64.manylinux_2_17_aarch64.manylinux_2_28_aarch64.whl.metadata (36 kB)
Collecting idna<4,>=2.5 (from requests->transformers)
Using cached idna-3.10-py3-none-any.whl.metadata (10 kB)
Collecting urllib3<3,>=1.21.1 (from requests->transformers)
Using cached urllib3-2.5.0-py3-none-any.whl.metadata (6.5 kB)
Collecting certifi>=2017.4.17 (from requests->transformers)
Using cached certifi-2025.8.3-py3-none-any.whl.metadata (2.4 kB)
Collecting aiohappyeyeballs>=2.5.0 (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.9.0,>=2023.1.0->datasets)
Downloading aiohappyeyeballs-2.6.1-py3-none-any.whl.metadata (5.9 kB)
Collecting aiosignal>=1.4.0 (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.9.0,>=2023.1.0->datasets)
Downloading aiosignal-1.4.0-py3-none-any.whl.metadata (3.7 kB)
Collecting async-timeout<6.0,>=4.0 (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.9.0,>=2023.1.0->datasets)
Downloading async_timeout-5.0.1-py3-none-any.whl.metadata (5.1 kB)
Collecting attrs>=17.3.0 (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.9.0,>=2023.1.0->datasets)
Downloading attrs-25.3.0-py3-none-any.whl.metadata (10 kB)
Collecting frozenlist>=1.1.1 (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.9.0,>=2023.1.0->datasets)
Downloading frozenlist-1.7.0-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl.metadata (18 kB)
Collecting multidict<7.0,>=4.5 (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.9.0,>=2023.1.0->datasets)

[illegible]

[illegible]

```

    yar1-1.20.1
(nanogpt) ddong@gh-login03:~/workspace/nanoGPT> python data/shakespeare_char/prepare.py
length of dataset in characters: 1,115,394
all the unique characters:
!$&',-.3:;?ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
vocab size: 65
train has 1,003,854 tokens
val has 111,540 tokens
(nanogpt) ddong@gh-login03:~/workspace/nanoGPT> srun -A bcrc-dtai-gh --qos=bcrc-dtai-gh -p ghx4 -t 4:00:00 --nodes=1 --ntasks-per-node=1
--cpus-per-task=16 --gres=gpu:h100:1 --mem=256g --pty bash -i
srun: job 970855 queued and waiting for resources
srun: job 970855 has been allocated resources
(base) ddong@gh098:~/workspace/nanoGPT> module load cuda/12.6.1

Lmod is automatically replacing "gcc/11.4.0" with "gcc-native/12.3".

AWS OFI NCCL Plugin v1.6.0 loaded for CUDA 12.6.1
This provides network transport ONLY - NCCL library from PyTorch
Configured for Slingshot11/CXI interconnect
CUDA 12.6.1 loaded with compatible AWS OFI NCCL plugin v1.6.0
Plugin provides network transport - NCCL library from PyTorch
Use with python/miniforge3_pytorch/2.7.0 or equivalent for best compatibility

Activating Modules:
    1) cray-libsci/24.07.0      2) cray-mpich/8.1.30

(base) ddong@gh098:~/workspace/nanoGPT> module load gcc/11.4.0

Lmod is automatically replacing "gcc-native/12.3" with "gcc/11.4.0".

Inactive Modules:
    1) cray-libsci      2) cray-mpich

(base) ddong@gh098:~/workspace/nanoGPT> conda activate nanogpt
(nanogpt) ddong@gh098:~/workspace/nanoGPT> python train.py config/train_shakespeare_char.py
Overriding config with config/train_shakespeare_char.py:
# train a miniature character-level shakespeare model
# good for debugging and playing on macbooks and such

out_dir = 'out-shakespeare-char'
eval_interval = 250 # keep frequent because we'll overfit
eval_iters = 200
log_interval = 10 # don't print too too often

```

```
# we expect to overfit on this small dataset, so only save when val improves
always_save_checkpoint = False
```

```
wandb_log = False # override via command line if you like
wandb_project = 'shakespeare-char'
wandb_run_name = 'mini-gpt'
```

```
dataset = 'shakespeare_char'
gradient_accumulation_steps = 1
batch_size = 64
block_size = 256 # context of up to 256 previous characters
```

```
# baby GPT model :)
n_layer = 6
n_head = 6
n_embd = 384
dropout = 0.2
```

```
learning_rate = 1e-3 # with baby networks can afford to go a bit higher
max_iters = 5000
lr_decay_iters = 5000 # make equal to max_iters usually
min_lr = 1e-4 # learning_rate / 10 usually
beta2 = 0.99 # make a bit bigger because number of tokens per iter is small
```

```
warmup_iters = 100 # not super necessary potentially
```

```
# on macbook also add
# device = 'cpu' # run on cpu only
compile = False # do not torch compile the model
```

```
tokens per iteration will be: 16,384
found vocab_size = 65 (inside data/shakespeare_char/meta.pkl)
Initializing a new model from scratch
number of parameters: 10.65M
```

```
/u/ddong/workspace/nanoGPT/train.py:196: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use
`torch.amp.GradScaler('cuda', args...)` instead.
```

```
scaler = torch.cuda.amp.GradScaler(enabled=(dtype == 'float16'))
num decayed parameter tensors: 26, with 10,740,096 parameters
num non-decayed parameter tensors: 13, with 4,992 parameters
using fused AdamW: True
step 0: train loss 4.2874, val loss 4.2823
iter 0: loss 4.2714, time 3520.31ms, mfu -100.00%
iter 10: loss 3.1476, time 12.42ms, mfu 30.00%
iter 20: loss 2.7333, time 12.17ms, mfu 30.06%
```

iter 30: loss 2.6157, time 12.02ms, mfu 30.15%
iter 40: loss 2.5761, time 12.08ms, mfu 30.22%
iter 50: loss 2.5257, time 12.19ms, mfu 30.26%
iter 60: loss 2.5101, time 12.18ms, mfu 30.29%
iter 70: loss 2.4937, time 12.17ms, mfu 30.33%
iter 80: loss 2.4949, time 12.14ms, mfu 30.36%
iter 90: loss 2.4642, time 12.29ms, mfu 30.36%
iter 100: loss 2.4559, time 12.06ms, mfu 30.41%
iter 110: loss 2.4545, time 12.18ms, mfu 30.43%
iter 120: loss 2.4294, time 12.05ms, mfu 30.48%
iter 130: loss 2.4133, time 12.24ms, mfu 30.48%
iter 140: loss 2.4042, time 12.19ms, mfu 30.49%
iter 150: loss 2.4085, time 12.06ms, mfu 30.53%
iter 160: loss 2.3676, time 12.12ms, mfu 30.55%
iter 170: loss 2.3564, time 12.33ms, mfu 30.52%
iter 180: loss 2.3013, time 12.19ms, mfu 30.52%
iter 190: loss 2.2393, time 12.17ms, mfu 30.53%
iter 200: loss 2.2029, time 11.99ms, mfu 30.59%
iter 210: loss 2.1412, time 12.04ms, mfu 30.62%
iter 220: loss 2.1340, time 12.06ms, mfu 30.65%
iter 230: loss 2.0779, time 12.15ms, mfu 30.65%
iter 240: loss 2.0766, time 12.06ms, mfu 30.68%
step 250: train loss 1.9638, val loss 2.0642
saving checkpoint to out-shakespeare-char
iter 250: loss 2.0311, time 2550.40ms, mfu 27.62%
iter 260: loss 1.9738, time 12.18ms, mfu 27.92%
iter 270: loss 1.9675, time 12.07ms, mfu 28.22%
iter 280: loss 1.9770, time 12.16ms, mfu 28.46%
iter 290: loss 1.9155, time 12.28ms, mfu 28.65%
iter 300: loss 1.8986, time 12.23ms, mfu 28.83%
iter 310: loss 1.8706, time 12.10ms, mfu 29.02%
iter 320: loss 1.8553, time 12.08ms, mfu 29.21%
iter 330: loss 1.8150, time 12.08ms, mfu 29.37%
iter 340: loss 1.7909, time 12.08ms, mfu 29.52%
iter 350: loss 1.8252, time 12.64ms, mfu 29.52%
iter 360: loss 1.7673, time 12.44ms, mfu 29.56%
iter 370: loss 1.7454, time 12.31ms, mfu 29.63%
iter 380: loss 1.7260, time 12.09ms, mfu 29.75%
iter 390: loss 1.7299, time 12.19ms, mfu 29.83%
iter 400: loss 1.7630, time 12.09ms, mfu 29.93%
iter 410: loss 1.6932, time 12.27ms, mfu 29.97%
iter 420: loss 1.7105, time 12.21ms, mfu 30.03%
iter 430: loss 1.6837, time 12.24ms, mfu 30.07%
iter 440: loss 1.6559, time 12.30ms, mfu 30.09%

iter 450: loss 1.6497, time 12.18ms, mfu 30.14%
iter 460: loss 1.5936, time 12.04ms, mfu 30.22%
iter 470: loss 1.6571, time 12.12ms, mfu 30.27%
iter 480: loss 1.6179, time 12.09ms, mfu 30.33%
iter 490: loss 1.5913, time 12.30ms, mfu 30.33%
step 500: train loss 1.5217, val loss 1.7301
saving checkpoint to out-shakespeare-char
iter 500: loss 1.6004, time 3461.56ms, mfu 27.31%
iter 510: loss 1.6053, time 12.50ms, mfu 27.56%
iter 520: loss 1.5869, time 12.23ms, mfu 27.85%
iter 530: loss 1.5560, time 12.26ms, mfu 28.10%
iter 540: loss 1.6128, time 12.18ms, mfu 28.35%
iter 550: loss 1.5617, time 12.69ms, mfu 28.45%
iter 560: loss 1.5615, time 12.19ms, mfu 28.66%
iter 570: loss 1.5583, time 12.43ms, mfu 28.79%
iter 580: loss 1.5294, time 12.08ms, mfu 29.00%
iter 590: loss 1.4889, time 12.39ms, mfu 29.11%
iter 600: loss 1.5138, time 12.60ms, mfu 29.15%
iter 610: loss 1.5298, time 12.41ms, mfu 29.24%
iter 620: loss 1.5307, time 12.17ms, mfu 29.38%
iter 630: loss 1.5098, time 12.35ms, mfu 29.46%
iter 640: loss 1.4556, time 12.11ms, mfu 29.59%
iter 650: loss 1.4932, time 12.20ms, mfu 29.69%
iter 660: loss 1.5016, time 12.46ms, mfu 29.71%
iter 670: loss 1.4396, time 12.29ms, mfu 29.77%
iter 680: loss 1.5061, time 12.05ms, mfu 29.88%
iter 690: loss 1.4545, time 11.95ms, mfu 30.02%
iter 700: loss 1.4737, time 12.20ms, mfu 30.07%
iter 710: loss 1.4501, time 12.01ms, mfu 30.16%
iter 720: loss 1.4389, time 12.25ms, mfu 30.19%
iter 730: loss 1.4234, time 12.07ms, mfu 30.26%
iter 740: loss 1.4216, time 12.05ms, mfu 30.32%
step 750: train loss 1.3575, val loss 1.5861
saving checkpoint to out-shakespeare-char
iter 750: loss 1.4249, time 2676.89ms, mfu 27.30%
iter 760: loss 1.4405, time 12.06ms, mfu 27.66%
iter 770: loss 1.4230, time 12.26ms, mfu 27.94%
iter 780: loss 1.3986, time 12.36ms, mfu 28.16%
iter 790: loss 1.4194, time 12.10ms, mfu 28.42%
iter 800: loss 1.4244, time 12.03ms, mfu 28.68%
iter 810: loss 1.4000, time 12.07ms, mfu 28.90%
iter 820: loss 1.4121, time 12.02ms, mfu 29.11%
iter 830: loss 1.3884, time 12.18ms, mfu 29.25%
iter 840: loss 1.3984, time 12.23ms, mfu 29.38%

iter 850: loss 1.3865, time 12.03ms, mfu 29.54%
iter 860: loss 1.3926, time 12.17ms, mfu 29.65%
iter 870: loss 1.3907, time 12.09ms, mfu 29.76%
iter 880: loss 1.3758, time 12.05ms, mfu 29.88%
iter 890: loss 1.3853, time 12.13ms, mfu 29.96%
iter 900: loss 1.3644, time 12.23ms, mfu 30.01%
iter 910: loss 1.3147, time 12.28ms, mfu 30.05%
iter 920: loss 1.3592, time 12.08ms, mfu 30.13%
iter 930: loss 1.3596, time 12.35ms, mfu 30.13%
iter 940: loss 1.3404, time 12.08ms, mfu 30.20%
iter 950: loss 1.3488, time 12.08ms, mfu 30.27%
iter 960: loss 1.3619, time 12.25ms, mfu 30.28%
iter 970: loss 1.3582, time 12.05ms, mfu 30.35%
iter 980: loss 1.3461, time 12.02ms, mfu 30.41%
iter 990: loss 1.3352, time 12.10ms, mfu 30.45%
step 1000: train loss 1.2701, val loss 1.5203
saving checkpoint to out-shakespeare-char
iter 1000: loss 1.3351, time 2646.91ms, mfu 27.42%
iter 1010: loss 1.3384, time 12.41ms, mfu 27.68%
iter 1020: loss 1.3077, time 12.27ms, mfu 27.95%
iter 1030: loss 1.3355, time 12.21ms, mfu 28.20%
iter 1040: loss 1.3564, time 12.40ms, mfu 28.39%
iter 1050: loss 1.2869, time 12.32ms, mfu 28.57%
iter 1060: loss 1.3379, time 12.66ms, mfu 28.66%
iter 1070: loss 1.3342, time 12.15ms, mfu 28.86%
iter 1080: loss 1.3318, time 12.06ms, mfu 29.07%
iter 1090: loss 1.3464, time 12.20ms, mfu 29.21%
iter 1100: loss 1.3153, time 12.00ms, mfu 29.40%
iter 1110: loss 1.2979, time 12.09ms, mfu 29.54%
iter 1120: loss 1.3029, time 12.31ms, mfu 29.61%
iter 1130: loss 1.2924, time 12.15ms, mfu 29.72%
iter 1140: loss 1.2999, time 12.15ms, mfu 29.81%
iter 1150: loss 1.3020, time 11.97ms, mfu 29.94%
iter 1160: loss 1.3272, time 12.15ms, mfu 30.02%
iter 1170: loss 1.2986, time 12.16ms, mfu 30.08%
iter 1180: loss 1.3087, time 12.06ms, mfu 30.16%
iter 1190: loss 1.2608, time 12.30ms, mfu 30.18%
iter 1200: loss 1.2894, time 12.07ms, mfu 30.25%
iter 1210: loss 1.2652, time 12.20ms, mfu 30.28%
iter 1220: loss 1.3068, time 12.10ms, mfu 30.33%
iter 1230: loss 1.2998, time 11.99ms, mfu 30.40%
iter 1240: loss 1.2930, time 12.31ms, mfu 30.39%
step 1250: train loss 1.2017, val loss 1.4890
saving checkpoint to out-shakespeare-char


```
iter 1250: loss 1.2713, time 2853.62ms, mfu 27.36%
iter 1260: loss 1.2832, time 12.22ms, mfu 27.68%
iter 1270: loss 1.2678, time 12.11ms, mfu 27.99%
iter 1280: loss 1.2520, time 12.16ms, mfu 28.25%
iter 1290: loss 1.2852, time 12.26ms, mfu 28.47%
iter 1300: loss 1.2956, time 12.15ms, mfu 28.68%
iter 1310: loss 1.2377, time 12.20ms, mfu 28.87%
iter 1320: loss 1.3061, time 12.04ms, mfu 29.08%
iter 1330: loss 1.2633, time 12.32ms, mfu 29.19%
iter 1340: loss 1.3004, time 12.07ms, mfu 29.36%
iter 1350: loss 1.2593, time 12.56ms, mfu 29.39%
iter 1360: loss 1.2744, time 12.41ms, mfu 29.46%
iter 1370: loss 1.2542, time 12.29ms, mfu 29.54%
iter 1380: loss 1.2582, time 12.16ms, mfu 29.65%
iter 1390: loss 1.2509, time 12.13ms, mfu 29.76%
iter 1400: loss 1.2585, time 12.00ms, mfu 29.89%
iter 1410: loss 1.2562, time 12.13ms, mfu 29.97%
iter 1420: loss 1.2705, time 12.03ms, mfu 30.07%
iter 1430: loss 1.2427, time 12.24ms, mfu 30.11%
iter 1440: loss 1.2563, time 12.15ms, mfu 30.17%
iter 1450: loss 1.2297, time 12.40ms, mfu 30.15%
iter 1460: loss 1.2486, time 12.19ms, mfu 30.20%
iter 1470: loss 1.2212, time 11.99ms, mfu 30.28%
iter 1480: loss 1.2117, time 12.13ms, mfu 30.33%
iter 1490: loss 1.2360, time 12.29ms, mfu 30.33%
step 1500: train loss 1.1531, val loss 1.4814
saving checkpoint to out-shakespeare-char
iter 1500: loss 1.1864, time 2658.29ms, mfu 27.31%
iter 1510: loss 1.2345, time 11.84ms, mfu 27.73%
iter 1520: loss 1.2195, time 12.28ms, mfu 27.99%
iter 1530: loss 1.2561, time 12.07ms, mfu 28.28%
iter 1540: loss 1.1985, time 12.22ms, mfu 28.50%
iter 1550: loss 1.2332, time 12.33ms, mfu 28.67%
iter 1560: loss 1.2111, time 12.02ms, mfu 28.90%
iter 1570: loss 1.2372, time 12.14ms, mfu 29.08%
iter 1580: loss 1.2026, time 12.18ms, mfu 29.23%
iter 1590: loss 1.1916, time 12.23ms, mfu 29.36%
iter 1600: loss 1.1960, time 12.13ms, mfu 29.49%
iter 1610: loss 1.2375, time 12.23ms, mfu 29.59%
iter 1620: loss 1.1826, time 12.00ms, mfu 29.73%
iter 1630: loss 1.2038, time 12.36ms, mfu 29.78%
iter 1640: loss 1.2102, time 12.12ms, mfu 29.87%
iter 1650: loss 1.1783, time 11.97ms, mfu 30.00%
iter 1660: loss 1.2219, time 12.25ms, mfu 30.04%
```

iter 1670: loss 1.1982, time 12.27ms, mfu 30.07%
iter 1680: loss 1.2076, time 12.12ms, mfu 30.14%
iter 1690: loss 1.2083, time 12.11ms, mfu 30.20%
iter 1700: loss 1.1900, time 12.21ms, mfu 30.23%
iter 1710: loss 1.1793, time 12.13ms, mfu 30.28%
iter 1720: loss 1.1819, time 12.14ms, mfu 30.33%
iter 1730: loss 1.2017, time 12.13ms, mfu 30.37%
iter 1740: loss 1.1737, time 12.17ms, mfu 30.39%
step 1750: train loss 1.1022, val loss 1.4627
saving checkpoint to out-shakespeare-char
iter 1750: loss 1.1886, time 2645.20ms, mfu 27.37%
iter 1760: loss 1.1886, time 12.28ms, mfu 27.66%
iter 1770: loss 1.2006, time 12.16ms, mfu 27.96%
iter 1780: loss 1.1916, time 12.22ms, mfu 28.21%
iter 1790: loss 1.1879, time 11.90ms, mfu 28.52%
iter 1800: loss 1.1874, time 12.01ms, mfu 28.77%
iter 1810: loss 1.1622, time 12.17ms, mfu 28.96%
iter 1820: loss 1.1666, time 11.88ms, mfu 29.20%
iter 1830: loss 1.1691, time 12.06ms, mfu 29.37%
iter 1840: loss 1.1635, time 12.19ms, mfu 29.49%
iter 1850: loss 1.1583, time 12.07ms, mfu 29.63%
iter 1860: loss 1.1767, time 12.23ms, mfu 29.71%
iter 1870: loss 1.1436, time 12.13ms, mfu 29.81%
iter 1880: loss 1.1821, time 12.12ms, mfu 29.90%
iter 1890: loss 1.1803, time 12.11ms, mfu 29.99%
iter 1900: loss 1.1353, time 12.17ms, mfu 30.05%
iter 1910: loss 1.1756, time 12.17ms, mfu 30.11%
iter 1920: loss 1.1753, time 12.15ms, mfu 30.17%
iter 1930: loss 1.1433, time 12.21ms, mfu 30.20%
iter 1940: loss 1.1303, time 12.19ms, mfu 30.24%
iter 1950: loss 1.1413, time 12.18ms, mfu 30.27%
iter 1960: loss 1.1558, time 12.14ms, mfu 30.31%
iter 1970: loss 1.1513, time 11.92ms, mfu 30.41%
iter 1980: loss 1.1545, time 12.06ms, mfu 30.46%
iter 1990: loss 1.1500, time 11.95ms, mfu 30.53%
step 2000: train loss 1.0584, val loss 1.4757
iter 2000: loss 1.1270, time 3071.19ms, mfu 27.49%
iter 2010: loss 1.1278, time 12.45ms, mfu 27.73%
iter 2020: loss 1.1233, time 12.12ms, mfu 28.03%
iter 2030: loss 1.1602, time 12.07ms, mfu 28.32%
iter 2040: loss 1.1431, time 12.33ms, mfu 28.51%
iter 2050: loss 1.1164, time 12.28ms, mfu 28.69%
iter 2060: loss 1.1041, time 12.19ms, mfu 28.88%
iter 2070: loss 1.1329, time 12.93ms, mfu 28.87%

iter 2080: loss 1.1266, time 12.20ms, mfu 29.04%
iter 2090: loss 1.1336, time 12.20ms, mfu 29.19%
iter 2100: loss 1.1306, time 11.96ms, mfu 29.39%
iter 2110: loss 1.1363, time 12.32ms, mfu 29.47%
iter 2120: loss 1.1350, time 12.37ms, mfu 29.54%
iter 2130: loss 1.1387, time 12.12ms, mfu 29.66%
iter 2140: loss 1.1427, time 12.00ms, mfu 29.80%
iter 2150: loss 1.1231, time 12.05ms, mfu 29.91%
iter 2160: loss 1.1400, time 12.14ms, mfu 29.99%
iter 2170: loss 1.1330, time 12.15ms, mfu 30.06%
iter 2180: loss 1.1149, time 12.19ms, mfu 30.11%
iter 2190: loss 1.1073, time 12.32ms, mfu 30.12%
iter 2200: loss 1.1294, time 12.01ms, mfu 30.21%
iter 2210: loss 1.1243, time 12.07ms, mfu 30.28%
iter 2220: loss 1.1295, time 12.23ms, mfu 30.30%
iter 2230: loss 1.1209, time 11.92ms, mfu 30.39%
iter 2240: loss 1.1328, time 11.97ms, mfu 30.46%
step 2250: train loss 1.0128, val loss 1.4754
iter 2250: loss 1.1120, time 3478.52ms, mfu 27.43%
iter 2260: loss 1.1086, time 12.19ms, mfu 27.74%
iter 2270: loss 1.1322, time 12.00ms, mfu 28.08%
iter 2280: loss 1.0989, time 12.10ms, mfu 28.35%
iter 2290: loss 1.1422, time 12.31ms, mfu 28.54%
iter 2300: loss 1.1258, time 12.44ms, mfu 28.68%
iter 2310: loss 1.0948, time 12.20ms, mfu 28.87%
iter 2320: loss 1.0992, time 12.15ms, mfu 29.05%
iter 2330: loss 1.1005, time 11.91ms, mfu 29.27%
iter 2340: loss 1.1191, time 12.04ms, mfu 29.44%
iter 2350: loss 1.1102, time 12.44ms, mfu 29.49%
iter 2360: loss 1.1090, time 12.21ms, mfu 29.59%
iter 2370: loss 1.0886, time 12.21ms, mfu 29.68%
iter 2380: loss 1.0815, time 12.36ms, mfu 29.73%
iter 2390: loss 1.0812, time 12.26ms, mfu 29.79%
iter 2400: loss 1.0873, time 12.08ms, mfu 29.90%
iter 2410: loss 1.0737, time 12.17ms, mfu 29.97%
iter 2420: loss 1.0838, time 12.15ms, mfu 30.04%
iter 2430: loss 1.0561, time 11.96ms, mfu 30.15%
iter 2440: loss 1.0624, time 12.17ms, mfu 30.20%
iter 2450: loss 1.0796, time 12.14ms, mfu 30.25%
iter 2460: loss 1.0866, time 12.21ms, mfu 30.27%
iter 2470: loss 1.0901, time 12.20ms, mfu 30.30%
iter 2480: loss 1.0835, time 12.04ms, mfu 30.37%
iter 2490: loss 1.0634, time 11.94ms, mfu 30.45%
step 2500: train loss 0.9638, val loss 1.4944

iter 2500: loss 1.0866, time 2205.75ms, mfu 27.42%
iter 2510: loss 1.0713, time 12.18ms, mfu 27.74%
iter 2520: loss 1.0542, time 12.39ms, mfu 27.97%
iter 2530: loss 1.0554, time 12.16ms, mfu 28.24%
iter 2540: loss 1.0608, time 12.10ms, mfu 28.49%
iter 2550: loss 1.0718, time 12.16ms, mfu 28.71%
iter 2560: loss 1.0654, time 12.18ms, mfu 28.90%
iter 2570: loss 1.0835, time 12.24ms, mfu 29.05%
iter 2580: loss 1.0774, time 12.61ms, mfu 29.10%
iter 2590: loss 1.0751, time 12.15ms, mfu 29.26%
iter 2600: loss 1.0672, time 12.32ms, mfu 29.36%
iter 2610: loss 1.0510, time 12.00ms, mfu 29.53%
iter 2620: loss 1.0540, time 12.31ms, mfu 29.60%
iter 2630: loss 1.0303, time 12.26ms, mfu 29.68%
iter 2640: loss 1.0493, time 16.88ms, mfu 28.92%
iter 2650: loss 1.0732, time 12.75ms, mfu 28.95%
iter 2660: loss 1.0387, time 12.60ms, mfu 29.01%
iter 2670: loss 1.0202, time 14.19ms, mfu 28.74%
iter 2680: loss 1.0497, time 13.75ms, mfu 28.57%
iter 2690: loss 1.0600, time 12.27ms, mfu 28.75%
iter 2700: loss 1.0259, time 13.23ms, mfu 28.69%
iter 2710: loss 1.0484, time 12.00ms, mfu 28.93%
iter 2720: loss 1.0453, time 227.80ms, mfu 26.20%
iter 2730: loss 1.0611, time 12.12ms, mfu 26.66%
iter 2740: loss 1.0280, time 12.59ms, mfu 26.95%
step 2750: train loss 0.9187, val loss 1.5082
iter 2750: loss 1.0385, time 3494.41ms, mfu 24.26%
iter 2760: loss 1.0404, time 12.22ms, mfu 24.89%
iter 2770: loss 1.0268, time 12.23ms, mfu 25.45%
iter 2780: loss 1.0290, time 12.00ms, mfu 26.01%
iter 2790: loss 1.0430, time 12.25ms, mfu 26.45%
iter 2800: loss 1.0224, time 12.10ms, mfu 26.88%
iter 2810: loss 1.0447, time 12.19ms, mfu 27.25%
iter 2820: loss 1.0264, time 12.50ms, mfu 27.51%
iter 2830: loss 1.0376, time 12.71ms, mfu 27.69%
iter 2840: loss 0.9989, time 443.47ms, mfu 25.00%
iter 2850: loss 1.0273, time 12.35ms, mfu 25.52%
iter 2860: loss 1.0281, time 12.07ms, mfu 26.06%
iter 2870: loss 1.0123, time 12.06ms, mfu 26.54%
iter 2880: loss 1.0382, time 12.15ms, mfu 26.95%
iter 2890: loss 1.0160, time 12.26ms, mfu 27.30%
iter 2900: loss 0.9958, time 12.03ms, mfu 27.66%
iter 2910: loss 1.0365, time 12.91ms, mfu 27.78%
iter 2920: loss 1.0136, time 12.38ms, mfu 28.02%

iter 2930: loss 1.0008, time 12.22ms, mfu 28.26%
iter 2940: loss 0.9884, time 12.05ms, mfu 28.53%
iter 2950: loss 1.0302, time 12.34ms, mfu 28.70%
iter 2960: loss 1.0023, time 12.86ms, mfu 28.72%
iter 2970: loss 0.9962, time 12.22ms, mfu 28.90%
iter 2980: loss 1.0050, time 12.07ms, mfu 29.10%
iter 2990: loss 0.9859, time 12.17ms, mfu 29.25%
step 3000: train loss 0.8727, val loss 1.5223
iter 3000: loss 0.9836, time 3097.46ms, mfu 26.34%
iter 3010: loss 0.9921, time 12.23ms, mfu 26.75%
iter 3020: loss 1.0038, time 12.15ms, mfu 27.14%
iter 3030: loss 1.0139, time 12.27ms, mfu 27.47%
iter 3040: loss 1.0366, time 12.31ms, mfu 27.75%
iter 3050: loss 0.9810, time 12.28ms, mfu 28.01%
iter 3060: loss 1.0031, time 12.29ms, mfu 28.24%
iter 3070: loss 1.0194, time 12.07ms, mfu 28.50%
iter 3080: loss 0.9975, time 12.36ms, mfu 28.67%
iter 3090: loss 0.9894, time 12.17ms, mfu 28.86%
iter 3100: loss 0.9978, time 12.10ms, mfu 29.06%
iter 3110: loss 0.9765, time 12.61ms, mfu 29.10%
iter 3120: loss 1.0015, time 12.26ms, mfu 29.23%
iter 3130: loss 0.9865, time 12.20ms, mfu 29.36%
iter 3140: loss 0.9873, time 11.96ms, mfu 29.54%
iter 3150: loss 1.0015, time 12.06ms, mfu 29.68%
iter 3160: loss 1.0062, time 12.19ms, mfu 29.77%
iter 3170: loss 0.9674, time 12.12ms, mfu 29.86%
iter 3180: loss 0.9737, time 12.17ms, mfu 29.94%
iter 3190: loss 1.0044, time 12.17ms, mfu 30.01%
iter 3200: loss 0.9637, time 12.20ms, mfu 30.06%
iter 3210: loss 0.9727, time 12.27ms, mfu 30.09%
iter 3220: loss 0.9645, time 12.00ms, mfu 30.19%
iter 3230: loss 0.9654, time 12.17ms, mfu 30.23%
iter 3240: loss 0.9645, time 12.06ms, mfu 30.30%
step 3250: train loss 0.8285, val loss 1.5549
iter 3250: loss 0.9885, time 3063.21ms, mfu 27.28%
iter 3260: loss 0.9630, time 12.09ms, mfu 27.63%
iter 3270: loss 0.9787, time 12.07ms, mfu 27.96%
iter 3280: loss 0.9547, time 12.03ms, mfu 28.26%
iter 3290: loss 0.9467, time 12.12ms, mfu 28.51%
iter 3300: loss 0.9528, time 12.36ms, mfu 28.67%
iter 3310: loss 0.9588, time 12.15ms, mfu 28.87%
iter 3320: loss 0.9751, time 12.50ms, mfu 28.96%
iter 3330: loss 0.9616, time 12.15ms, mfu 29.14%
iter 3340: loss 0.9526, time 12.50ms, mfu 29.20%

iter 3350: loss 0.9598, time 12.01ms, mfu 29.38%
iter 3360: loss 0.9378, time 12.01ms, mfu 29.55%
iter 3370: loss 0.9615, time 12.34ms, mfu 29.61%
iter 3380: loss 0.9416, time 12.33ms, mfu 29.67%
iter 3390: loss 0.9555, time 12.12ms, mfu 29.78%
iter 3400: loss 0.9631, time 12.46ms, mfu 29.79%
iter 3410: loss 0.9544, time 12.48ms, mfu 29.80%
iter 3420: loss 0.9642, time 12.17ms, mfu 29.88%
iter 3430: loss 0.9422, time 12.10ms, mfu 29.97%
iter 3440: loss 0.9860, time 12.62ms, mfu 29.93%
iter 3450: loss 0.9636, time 12.21ms, mfu 29.98%
iter 3460: loss 0.9549, time 12.45ms, mfu 29.98%
iter 3470: loss 0.9454, time 12.51ms, mfu 29.96%
iter 3480: loss 0.9575, time 12.46ms, mfu 29.95%
iter 3490: loss 0.9279, time 12.13ms, mfu 30.03%
step 3500: train loss 0.7873, val loss 1.5700
iter 3500: loss 0.9083, time 2436.86ms, mfu 27.04%
iter 3510: loss 0.9217, time 12.15ms, mfu 27.41%
iter 3520: loss 0.9299, time 12.37ms, mfu 27.68%
iter 3530: loss 0.9566, time 12.19ms, mfu 27.97%
iter 3540: loss 0.9342, time 12.03ms, mfu 28.27%
iter 3550: loss 0.9330, time 12.54ms, mfu 28.41%
iter 3560: loss 0.9595, time 13.15ms, mfu 28.41%
iter 3570: loss 0.9409, time 12.21ms, mfu 28.62%
iter 3580: loss 0.9452, time 12.76ms, mfu 28.68%
iter 3590: loss 0.9270, time 12.07ms, mfu 28.89%
iter 3600: loss 0.9325, time 12.10ms, mfu 29.08%
iter 3610: loss 0.9188, time 12.09ms, mfu 29.26%
iter 3620: loss 0.9191, time 12.28ms, mfu 29.36%
iter 3630: loss 0.9318, time 12.15ms, mfu 29.50%
iter 3640: loss 0.9185, time 12.34ms, mfu 29.57%
iter 3650: loss 0.9152, time 12.24ms, mfu 29.65%
iter 3660: loss 0.9518, time 12.15ms, mfu 29.75%
iter 3670: loss 0.9445, time 12.16ms, mfu 29.84%
iter 3680: loss 0.9093, time 12.30ms, mfu 29.89%
iter 3690: loss 0.9440, time 12.15ms, mfu 29.97%
iter 3700: loss 0.8836, time 12.13ms, mfu 30.04%
iter 3710: loss 0.8875, time 12.22ms, mfu 30.09%
iter 3720: loss 0.9176, time 12.24ms, mfu 30.12%
iter 3730: loss 0.9091, time 12.05ms, mfu 30.20%
iter 3740: loss 0.9067, time 12.14ms, mfu 30.25%
step 3750: train loss 0.7509, val loss 1.5937
iter 3750: loss 0.8996, time 2644.23ms, mfu 27.24%
iter 3760: loss 0.9355, time 12.30ms, mfu 27.54%

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iter 3770: loss 0.9404, time 12.08ms, mfu 27.87%
iter 3780: loss 0.9230, time 12.06ms, mfu 28.18%
iter 3790: loss 0.8987, time 12.10ms, mfu 28.44%
iter 3800: loss 0.9212, time 12.93ms, mfu 28.48%
iter 3810: loss 0.9212, time 12.04ms, mfu 28.72%
iter 3820: loss 0.8863, time 12.21ms, mfu 28.90%
iter 3830: loss 0.9080, time 12.72ms, mfu 28.94%
iter 3840: loss 0.9014, time 12.24ms, mfu 29.09%
iter 3850: loss 0.8957, time 12.21ms, mfu 29.24%
iter 3860: loss 0.8774, time 12.08ms, mfu 29.40%
iter 3870: loss 0.8987, time 12.75ms, mfu 29.38%
iter 3880: loss 0.8940, time 12.40ms, mfu 29.45%
iter 3890: loss 0.9065, time 12.57ms, mfu 29.47%
iter 3900: loss 0.8986, time 12.21ms, mfu 29.57%
iter 3910: loss 0.9005, time 12.08ms, mfu 29.70%
iter 3920: loss 0.8798, time 12.09ms, mfu 29.81%
iter 3930: loss 0.9021, time 12.09ms, mfu 29.91%
iter 3940: loss 0.8863, time 13.75ms, mfu 29.63%
iter 3950: loss 0.8870, time 12.17ms, mfu 29.73%
iter 3960: loss 0.9221, time 12.35ms, mfu 29.78%
iter 3970: loss 0.9039, time 12.12ms, mfu 29.87%
iter 3980: loss 0.9062, time 12.12ms, mfu 29.96%
iter 3990: loss 0.8865, time 11.93ms, mfu 30.09%
step 4000: train loss 0.7162, val loss 1.6193
iter 4000: loss 0.8655, time 2430.88ms, mfu 27.09%
iter 4010: loss 0.8844, time 12.16ms, mfu 27.45%
iter 4020: loss 0.8958, time 11.93ms, mfu 27.83%
iter 4030: loss 0.8803, time 12.06ms, mfu 28.13%
iter 4040: loss 0.8821, time 12.10ms, mfu 28.40%
iter 4050: loss 0.8759, time 12.15ms, mfu 28.63%
iter 4060: loss 0.8727, time 12.09ms, mfu 28.85%
iter 4070: loss 0.8732, time 12.12ms, mfu 29.04%
iter 4080: loss 0.8963, time 12.14ms, mfu 29.20%
iter 4090: loss 0.8644, time 12.12ms, mfu 29.36%
iter 4100: loss 0.9028, time 12.04ms, mfu 29.52%
iter 4110: loss 0.8807, time 12.12ms, mfu 29.64%
iter 4120: loss 0.8757, time 12.00ms, mfu 29.78%
iter 4130: loss 0.8669, time 12.15ms, mfu 29.87%
iter 4140: loss 0.8857, time 12.29ms, mfu 29.91%
iter 4150: loss 0.8732, time 12.31ms, mfu 29.95%
iter 4160: loss 0.8688, time 12.34ms, mfu 29.98%
iter 4170: loss 0.8780, time 12.01ms, mfu 30.08%
iter 4180: loss 0.8706, time 12.22ms, mfu 30.12%
iter 4190: loss 0.8694, time 12.06ms, mfu 30.20%
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iter 4200: loss 0.8581, time 12.19ms, mfu 30.23%
iter 4210: loss 0.8802, time 12.00ms, mfu 30.32%
iter 4220: loss 0.8666, time 12.17ms, mfu 30.35%
iter 4230: loss 0.8912, time 12.08ms, mfu 30.40%
iter 4240: loss 0.8704, time 12.03ms, mfu 30.45%
step 4250: train loss 0.6858, val loss 1.6435
iter 4250: loss 0.8708, time 2207.34ms, mfu 27.42%
iter 4260: loss 0.8613, time 12.21ms, mfu 27.73%
iter 4270: loss 0.8679, time 12.09ms, mfu 28.04%
iter 4280: loss 0.8560, time 12.06ms, mfu 28.33%
iter 4290: loss 0.8415, time 12.11ms, mfu 28.57%
iter 4300: loss 0.8258, time 12.19ms, mfu 28.77%
iter 4310: loss 0.8557, time 12.18ms, mfu 28.95%
iter 4320: loss 0.8447, time 11.96ms, mfu 29.17%
iter 4330: loss 0.8677, time 12.04ms, mfu 29.35%
iter 4340: loss 0.8396, time 12.07ms, mfu 29.50%
iter 4350: loss 0.8464, time 12.21ms, mfu 29.60%
iter 4360: loss 0.8685, time 12.19ms, mfu 29.70%
iter 4370: loss 0.8611, time 12.38ms, mfu 29.74%
iter 4380: loss 0.8477, time 11.91ms, mfu 29.89%
iter 4390: loss 0.8771, time 12.10ms, mfu 29.98%
iter 4400: loss 0.8614, time 12.08ms, mfu 30.07%
iter 4410: loss 0.8703, time 12.21ms, mfu 30.12%
iter 4420: loss 0.8690, time 12.23ms, mfu 30.15%
iter 4430: loss 0.8533, time 12.17ms, mfu 30.20%
iter 4440: loss 0.8524, time 12.18ms, mfu 30.24%
iter 4450: loss 0.8536, time 12.20ms, mfu 30.27%
iter 4460: loss 0.8397, time 12.14ms, mfu 30.31%
iter 4470: loss 0.8523, time 12.20ms, mfu 30.33%
iter 4480: loss 0.8429, time 11.96ms, mfu 30.42%
iter 4490: loss 0.8585, time 12.11ms, mfu 30.45%
step 4500: train loss 0.6629, val loss 1.6588
iter 4500: loss 0.8659, time 2216.48ms, mfu 27.42%
iter 4510: loss 0.8512, time 12.07ms, mfu 27.77%
iter 4520: loss 0.8441, time 12.01ms, mfu 28.09%
iter 4530: loss 0.8546, time 12.13ms, mfu 28.36%
iter 4540: loss 0.8531, time 12.21ms, mfu 28.57%
iter 4550: loss 0.8819, time 12.16ms, mfu 28.78%
iter 4560: loss 0.8351, time 12.13ms, mfu 28.97%
iter 4570: loss 0.8516, time 12.22ms, mfu 29.12%
iter 4580: loss 0.8632, time 11.99ms, mfu 29.32%
iter 4590: loss 0.8719, time 12.34ms, mfu 29.41%
iter 4600: loss 0.8269, time 12.33ms, mfu 29.49%
iter 4610: loss 0.8704, time 12.22ms, mfu 29.59%

iter 4620: loss 0.8474, time 12.24ms, mfu 29.68%
iter 4630: loss 0.8317, time 12.33ms, mfu 29.73%
iter 4640: loss 0.8508, time 12.45ms, mfu 29.75%
iter 4650: loss 0.8766, time 12.21ms, mfu 29.83%
iter 4660: loss 0.8533, time 12.50ms, mfu 29.82%
iter 4670: loss 0.8426, time 12.15ms, mfu 29.91%
iter 4680: loss 0.8577, time 12.32ms, mfu 29.94%
iter 4690: loss 0.8558, time 12.31ms, mfu 29.98%
iter 4700: loss 0.8331, time 12.17ms, mfu 30.04%
iter 4710: loss 0.8015, time 12.22ms, mfu 30.08%
iter 4720: loss 0.8335, time 12.40ms, mfu 30.08%
iter 4730: loss 0.8265, time 12.37ms, mfu 30.09%
iter 4740: loss 0.8421, time 12.27ms, mfu 30.12%
step 4750: train loss 0.6446, val loss 1.6769
iter 4750: loss 0.8188, time 2262.30ms, mfu 27.12%
iter 4760: loss 0.8216, time 12.49ms, mfu 27.39%
iter 4770: loss 0.8094, time 12.37ms, mfu 27.67%
iter 4780: loss 0.8205, time 12.40ms, mfu 27.90%
iter 4790: loss 0.8522, time 12.10ms, mfu 28.19%
iter 4800: loss 0.8315, time 12.38ms, mfu 28.38%
iter 4810: loss 0.8352, time 12.49ms, mfu 28.53%
iter 4820: loss 0.8244, time 12.34ms, mfu 28.69%
iter 4830: loss 0.8300, time 12.18ms, mfu 28.88%
iter 4840: loss 0.8442, time 12.40ms, mfu 29.00%
iter 4850: loss 0.8345, time 12.39ms, mfu 29.11%
iter 4860: loss 0.8237, time 12.03ms, mfu 29.29%
iter 4870: loss 0.8187, time 12.16ms, mfu 29.43%
iter 4880: loss 0.8363, time 12.19ms, mfu 29.54%
iter 4890: loss 0.8198, time 12.29ms, mfu 29.62%
iter 4900: loss 0.8156, time 12.33ms, mfu 29.68%
iter 4910: loss 0.8341, time 12.32ms, mfu 29.74%
iter 4920: loss 0.8262, time 12.44ms, mfu 29.76%
iter 4930: loss 0.8130, time 12.22ms, mfu 29.83%
iter 4940: loss 0.8096, time 12.42ms, mfu 29.85%
iter 4950: loss 0.8373, time 12.45ms, mfu 29.86%
iter 4960: loss 0.8353, time 12.52ms, mfu 29.85%
iter 4970: loss 0.7957, time 12.34ms, mfu 29.88%
iter 4980: loss 0.8014, time 12.24ms, mfu 29.94%
iter 4990: loss 0.8311, time 12.33ms, mfu 29.97%
step 5000: train loss 0.6296, val loss 1.6980
iter 5000: loss 0.8360, time 2260.00ms, mfu 26.99%

(nanogpt) ddong@gh-login03:~/workspace/nanoGPT> tmux capture-pane -p -e -J -S - -E - -> tmux-pane-color-\$(date +%F_%H%M%S).log